🌈 AS7343 Spectral Sensor

Hardware Overview

🌈 What is it?

The **AS7343 Spectral Sensor** is a light and color sensor. Unlike a normal camera that only sees Red, Green, and Blue, it can detect `14` specific wavelengths of light, ranging from invisible Ultraviolet (UV) to Near-Infrared (NIR). It is used to analyze the physical "fingerprint" of the light bouncing off an object.

⚡ Electrical Explanation

The sensor contains a grid of photodiodes. Over these diodes are optical filters (like tiny glass windows) that only let very specific colors of light pass through.

- **Auto-Multiplexing:** Because it has many "windows," the chip has to quickly switch its internal wiring back and forth to read all the colors.
- **The Flashlight:** The board also includes a bright white LED. By shining this pure white light onto an object and measuring what bounces back, you can figure out the object's color.

Key Hardware Facts:

- **Connection:** I²C Bus (`SDA`/`SCL`).
- **Default Address:** `0x39`.
- **Voltage:** Operates at **`3.3`V to `5`V**.
- **Signal Flow:** Light ➔ Optical Filters ➔ Photodiodes ➔ I²C Bus ➔ K-Comp Processor.

Software & Programming

📚 `kcomp_AS7343.h`

Because the sensor takes over `100` milliseconds to physically switch its internal wiring and read all `14` channels, reading it manually is a bit complicated. This library handles all the timing and I2C communication.

Usage: `#include <kcomp_AS7343.h>`

📦 The Data Structure

Unlike other sensors that give you a single number (like `int`), the AS7343 gives you a `struct` (a package of variables). When you call `getAS7343Data()`, you get a package containing all of this:

Field	Type	Description
`valid`	`bool`	`true` if the sensor successfully read data.
`clear`	`uint16_t`	The total unfiltered brightness of the light.
`f1_405nm, f2_425nm, fz_450nm`	`uint16_t`	Deep Violet and Blue light bands.
`f3_475nm, f4_515nm, f5_550nm`	`uint16_t`	Cyan and Green light bands.
`fy_555nm, fxl_600nm, f6_640nm`	`uint16_t`	Yellow, Orange, and Red light bands.
`f7_690nm, f8_745nm`	`uint16_t`	Deep Red and border-line infrared bands.
`nir`	`uint16_t`	**Near-Infrared** (invisible heat/light).

Function Reference

Function Name	Description & Parameters
`bool initAS7343()`	**Setup:** Wakes up the sensor and configures it. **Returns**: `true` if successful.
`bool updateAS7343()`	**The Worker:** Tells the sensor to scan the light. Takes about `~150ms`. Returns `true` when data is ready.
`AS7343Data getAS7343Data()`	Returns the package of spectral data described above.
`void setAS7343LED(bool enable, uint8_t mA)`	**Config:** Turns the onboard white LED on or off. ▪ `enable`: `true` or `false`. ▪ `mA`: Brightness from `4` to `10` (e.g., `4` is a safe default).
`void setAS7343Gain(uint8_t gain_level)`	**Config:** Changes how sensitive the sensor is to light. ▪ `gain_level`: `0` to `12`. (e.g., `6` is a suitable default).

⚠️ Pitfalls & Troubleshooting

- **The "Glare" Blindness**

  If you hold a glossy plastic object directly against the sensor, the white LED will reflect straight back into the lens. This blinds the sensor with and washes out the colors.

  The Fix: Hold objects at a slight angle, about **2 to 3 centimeters** above the sensor!

📝 Sample Program

This program turns on the LED and prints some spectral data.

```
#include <kcomp_AS7343.h>

void setup() {
  Serial.begin(115200);

  if (!initAS7343()) {
    Serial.println("❌ ERROR: Sensor not found. Are the wires plugged in?");
    while (1) { delay(100); }
  }

  setAS7343Gain(6);
  setAS7343LED(true, 12);
}

void loop() {
  if (updateAS7343()) {

    AS7343Data lightData = getAS7343Data();

    // The Brightness Reading
    Serial.print("◯ Total Brightness: "); Serial.println(lightData.clear);

    Serial.println("\n🔬 Raw Spectral Data (Yellow, Orange, & Red)");
    // Yellow, Orange, & Red
    Serial.print("555nm: "); Serial.print(lightData.fy_555nm); Serial.print("\t");
    Serial.print("600nm:"); Serial.print(lightData.fxl_600nm); Serial.print("\t");
    Serial.print("640nm: "); Serial.print(lightData.f6_640nm); Serial.println();
  } else {
    Serial.println("⚠ Oops! Couldn't read the sensor. Is a wire loose?");
  }
  delay(1000);
}
```

💡 **Try it out:** You can find the full sample program **`SimpleSpectralSensorAS7343`** in the K-Comp examples folder!